

**Project Design Phase-II
Technology Stack**

Date	03 October 2026
Team ID	SWTID-2026-6471
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	4 Marks

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	User interacts with EduGenie	HTML, CSS
2.	Application Logic-1	Handles application requests	Python, FastAPI
3.	Application Logic-2	Generates answers and quizzes	Google Gemini
4.	Application Logic-3	Provide simple explanations	LaMini-Flan-T5
5.	Database	Store application data.	Not used.

6.	Cloud Database	Database Service on Cloud	Not used
7.	File Storage	Stores project files	Local Filesystem
8.	External API-1	AI-based learning responses	Google Gemini API
9.	External API-2	Additional external API	Not used
10.	Machine Learning Model	Provides concept explanations	Lamini-Flan-T5-783M
11.	Infrastructure (Server / Cloud)	Run the application	Local server,Uvicorn

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used to build the application	FastAPI, Jinja2
2.	Security Implementations	API Key is stored securely	.env ,API key.
3.	Scalable Architecture	Separate modules for each feature	FastAPI, Modular design

4.	Availability	Application runs on a local server	Uvicorn
5.	Performance	Lightweight web interface and backend.	FastAPI, Gemini API